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Module 5

Data Analytics, Data Science & Visualization with Python (48 hours 16 Session)

Python Basics

- What is Python
- Application of Python
- Why use Python for AIML Installation
- Anaconda/Other Idle
- Python Tokens
- Data Types in Python
- Conditional Statement
- Loops in Python
- Functions in Python
- Advance Functions
- File Handling
- **Mini Project**

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Module 5

Intro to Git & Github and Kaggle

- Git and GitHub
- kaggle

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Module 5

Python NumPy & Pandas

- Introduction to NumPy
- Exploring a NumPy Array
- Indexing & Slicing a NumPy Array
- Manipulating a NumPy Array
- Performing Mathematical & Statistical Functions using NumPy
- Performing Linear Algebra Operations using NumPy
- Introduction to Pandas
- Exploring Pandas Series
- Introduction to Pandas DataFrame
- Importing & Exporting Data
- Implementing basic DataFrame functionalities
- Exploring Descriptive Statistics with Pandas
- **Mini Project**

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R Language Essentials

- Fundamentals of R
- Vectors & control statements
- Functions in R
- Matrices & strings
- Lists & arrays in R
- Data visualization in R

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Module 5

Exploratory Data Analysis (EDA)

- Data Types
- Dispersion & Skewness
- Data imputation
- Data Pre-processing
- Data Cleaning
- Data Manipulation
- Advanced Manipulation
- **Mini Project**

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SQL For Data Analytics

- SQL Basic
- SQL Joins
- SQL Aggregations
- Subqueries and Temp Tables
- SQL Data Cleaning
- Window Functions
- **Mini Project**

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Data Analysis with Excel

- Key Formulas and Functions, Ranges and Tables
- Data Cleaning – Text Functions, Dates and Times
- Conditional Formatting, Sorting and Filtering
- Dashboard Creation
- Analysis with Pivot Tables
- Data Analysis in Excel – Trends and Patterns
- Data Visualization in Excel – Charts and Plots
- Functions in Python
- Advance Functions File Handling
- **Mini Project**

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Data Visualization with Python

- Why Data Visualization?
- Introduction to Data Visualization
- Libraries & Tools for Data Visualization in Python
- Static Data Visualization Using Seaborn
- Interactive Data Visualization Using Plotly Express
- Interactive Animations & Facet Plots

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Tableau for Business Intelligence

- Understanding Data
- Creating Your First visualization
- Tableau Calculations
- Formatting Visualizations
- Manipulating Data in Tableau
- Creating Dashboards AND Stories
- Distributing & Publishing Your Visualization
- **Mini Project**

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Visual Storytelling using Power BI

- Introduction To Power BI
- Creating Power BI Reports, Auto Filters
- Report Visualization And Properties
- Chart And Map Report Properties
- Hierarchies And Drilldown Reports
- Power BI Deployment & Cloud
- Improving Power BI Reports And More...
- **Mini Project**

Capstone Project- 1 in Data Analytics

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Module 5

Applied Maths, Statistics and Probability for ML & AI (21 Hours 7 Sessions)

Mathematics for Machine Learning & AI

- Linear Algebra
- Multi-variable Calculus
- Introduction to Calculus

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Statistics for Machine Learning & AI

- Applications of Statistics
- Introduction to Statistics
- Categories of Data
- Basic Terminologies in Statistics
- Sampling techniques
- Descriptive Statistics
- Measure Used in Descriptive Statistics
- Z-Scores

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Probability

- What is Probability?
- Rules of probability
- Types of Probability
- Random Variables
- Probability Distribution Functions
- Mini Project

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Inferential Statistics

- Introduction to Inferential Statistics
- Hypothesis Testing
- Normal Distribution
- P-value
- One-tailed and Two-tailed tests
- One Sample Z test
- One Sample T test
- Independent Sample T test
- Chi-square test
- ANOVA

Capstone Project- 2

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Machine Learning Specialization (60 Hours 20 Sessions)

Introduction to Machine Learning

- What is Machine Learning?
- Applications of Machine Learning
- Machine Learning in your daily life
- Machine Learning in Retail
- Steps Involved in Machine Learning

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Regression

- Introduction to Regression
- Linear Regression
- Evaluation Metrics in Regression Models
- Logistic Regression
- **Mini Project**

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Module 5

Supervised Classification

- Why Use Classification?
- Application of Classification Algorithms
- Introduction to Classification
- Types of Classification Algorithms
- Classification: Decision Tree
- Classification: Random Forest
- ML in Banking & Finance – Benefits
- Classification: SVM
- Classification: KNN
- Classification: Naïve Bayes
- Evaluating Classification Models
- Model Optimization Techniques
- Model Boosting Techniques
- Introduction to PyCaret
- Dealing with Unbalanced Datasets
- **4 Mini Projects**

Capstone Project- 3

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Unsupervised Learning

- What is Unsupervised Learning?
- Application of Unsupervised Learning
- Introduction to Clustering
- Types of Clustering
- Partitioning Methods: K-means, DBSCAN, Spectral
- Hierarchical Methods: Hierarchical
- **2 Mini Projects**

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Dimension Reduction

- PCA
- Factor Analysis
- LDA
- **Mini Project**

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Association Rules Mining

- What are Association Rules?
- Association Rule Parameters
- A-priori Algorithm
- Market Basket Analysis

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Recommendation System

- What is a Recommendation System?
- Need for a Recommendation System
- Recommendation System Use Cases
- Applications of Recommendation System
- Types of Recommendation Systems
- Collaborative Filtering
- Content Based Filtering
- Matrix Factorization
- Pros and Cons of Collaborative Filtering
- Hybrid Recommender System
- **Mini Project**

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Module 5

Time-series Forecasting

- Introduction to forecasting data
- Properties of Time Series data
- Features of Time Series data
- Markov Processes – Overview and Terminologies
- Naive, Average and Moving Average Forecasting
- Exponential Smoothing
- ARIMA Approach
- **Mini Project**

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Machine Learning Model Deployment

- Overview of Machine Learning Models
- Machine Learning System Architecture
- Research Environment
- Packaging, Serving and Deploying the model
- Differential Testing And More...

Capstone Project- 4

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Deep Learning & AI Specialization (57 Hours 19 Sessions)

Introduction to Deep Learning

- What is Deep Learning
- Curse of Dimensionality
- Machine Learning vs. Deep Learning
- Use Cases of Deep Learning
- Human Brain vs. Neural Network
- What is Perceptron?
- Learning Rate
- Epoch
- Batch Size

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Tensorflow 2.0 with Tensor Board

- Introduction to Tensorflow 2.x
- Installing Tensorflow 2.x
- Introduction to TensorBoard
- Defining Sequence model layers
- Activation Function
- Layer Types
- Model Compilation
- Model Optimizer
- Model Loss Function
- Model Training
- Digit Classification using Simple Neural Network in Tensorflow 2.x
- **Mini Project**

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Computer Vision

- Introduction to Convolutional Neural Networks
- Introduction to images
- Convolution, Pooling, Padding & its mechanisms
- Forward Propagation & Backpropagation for CNNs
- CNN architectures
- AlexNet, VGGNet, InceptionNet & ResNet
- Transfer Learning
- Object Detection

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Introduction to NLP

- Introduction to NLP
- Libraries & Tools for NLP in Python
- NLTK vs Spacy
- Applications of NLP:
- Chatbot
- Search, Autocorrect and Autocomplete
- Grammar Checker

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Text Processing Methods

- Bag of Words
- Countvectorizer
- Term Frequency (TF)
- Inverse Document Frequency (IDF)
- Converting text to features and labels
- Multinomial Naive Bayes Classifier
- Leveraging Confusion Matrix Assignment
- Word Embeddings
- Word2Vec
- **Mini Project**

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TensorFlow Hub for Object Detection using Faster RCNN

- Introduction to TensorFlow Hub
- Use cases of TensorFlow Hub
- Limitations of CNN in object detection
- Architecture of RCNN
- Applications of RCNN
- Types of RCNN
- Step by step implementation of Faster RCNN

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Sentiment Analysis

- Sentiment Analysis
- Subjectivity Analysis
- Topic Extraction
- Product Reviews
- Opinion Retrieval and Spam
- Opinion Summarization
- Implementing Sentiment Analysis in Python

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Reinforcement Learning

- Understanding Reinforcement Learning
- Algorithms associated with RL
- Q-Learning Model
- **Mini Project**

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MLOps & Deployment

- Introduction To MLOps
- Automating Pipelines with Airflow
- Model Packaging
- API Building
- Model Training with AWS SageMaker
- Model Deployment
- Model Maintenance

Capstone Project- 5

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Advanced Generative AI, LLMs & Agentic AI Specialization (54 Hours 18 Sessions)

Introduction To Generative AI

- Introduction to Generative AI Models
- The Future of Generative AI
- Types of Generative AI Models
- Ethical Considerations in Generative AI Models & ChatGPT
- Popular Generative AI Models
- Benchmarking & Evaluating Models

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ChatGPT, Prompt Engineering, Explainable AI ChatGPT

Fundamentals of Prompt Engineering

- Zero-shot, Few-shot, Chain-of-Thought (CoT) Promptin
- Role-based Prompts, Multi-turn Conversations

Advanced Prompt Techniques

- System vs. User Prompts, Style Transfer, Iterative Prompting

Explainable AI

- Introduction to explainable AI
- Mini Project

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Advanced Generative AI Models & Architecture

- Introduction to Generative Models
- Large Language Models (LLMs)
- LLMs: GPT-4, DeepSeek, LLaMA, Mistral, Claude, Falcon
- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs) – StyleGAN, CycleGAN
- Attention Mechanisms and Transformers (GPT, BERT, T5)
- **Mini Project**

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Advanced Generative AI - Building LLM Applications

- Langchain and Workflow Design
- Advanced Prompt Engineering Techniques
- LLM Application Development
- LLM Fine-Tuning and Customization
- Fine-Tuning & Pre-training of LLMs
- Benchmarking and Evaluation of LLM Capabilities
- LangChain for LLM Development
- RAG with Hugging Face & LangChain
- **2 Mini Projects**

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Agentic AI & Multimodal AI (Voice, Video & Document AI)

Agentic AI

- Frameworks
- Hugging Face Ecosystem
- Speech-to-Text & AI Voice Processing
- Whisper, Wav2Vec, DeepSeek-STT

Multimodal AI Text-to-Image & Image-to-Image Generation

- DALL·E, MidJourney, and Stable Diffusion
- LDMs & Hugging Face Implementations

Advanced Multimodal Models

- Google's Gemini, OpenAI's GPT-4o, and models from Meta AI
- **Mini Project**

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Generative AI Applications In Industries

- Applications in Healthcare (e.g., Medical Imaging, Diagnosis)
- Applications in Finance (e.g., Risk Modeling, Fraud Detection)
- Applications in Retail and E-commerce (e.g., Personalized Marketing)
- Applications in Media and Entertainment (e.g., Content Creation)

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Responsible AI

- Awareness, Toxicity, Misinformation, Bias, Security, Fairness